1 The Succinctness Gap in Quantum Automata in a 1-Qubit Matrix Transformation Framework

Succinctness Gap in Quantum Automata

BS-CS-27-130

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Advisor: Dr. Jeremy Miller

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The goal of this project is to solve the succinctness problem in quantum formal language theory by investigating the 1-Qubit Quantum Finite Automaton (QFA) through the lens of linear algebraic transformations. While QFAs are known to require exponentially fewer states than classical deterministic automata for certain languages, finding the exact bounds of this succinctness gap remains a critical open challenge. By conceptualizing quantum operations as continuous state transitions driven by 2×2 unitary matrices, we establish a rigorous mathematical framework to analyze quantum state complexity. We demonstrate the step-by-step processing of binary strings to calculate probability amplitudes, utilizing this matrix-driven approach as a foundational step toward systematically characterizing the exponential memory efficiency of quantum automata.

Keywords: Quantum Finite Automata, Formal Language Theory, Succinctness Gap, Unitary Matrices, State Complexity.

2 Trading Eye: Game Theory Method for Optimizing Stock-Market Trading Strategies

Trading Eye: A Game Theory Method for Building Trading Strategies in the Live Stock Market.

BS-CS-27-131

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The goal of this project is to construct *Trading Eye*: an algorithmic trading platform tailored to design, backtest, and optimize real-world financial strategies. It scans financial markets for the first ten or more assets—stocks or currencies—that meet a defined long-entry signal. The user is free to build customized strategies based on the basic indicators including Simple, Exponential, and Hull Moving Averages (SMAs, EMAs, HMAs), Volatility, Bollinger bands, and more. Trading Eye should also accommodate delta-neutral portfolios based on the Black-Scholes equation. The app simulates and visualizes profit margins in real time or using historical data. It supports multiple strategies,

each based on probabilistic models trained on past market behavior, and allocates investments using weighted distributions. Users can compare the performance of arbitrary strategies via clear, interactive graphs. The system is designed to aid traders in identifying high-probability opportunities and optimizing portfolio decisions with data-driven insights.

Keywords: algorithmic trading, real-time graphs, historical data, probability-based strategy, weighted investments, profit comparison.

3 Building a Turing Machine that Computes the Genetic Translation of a DNA Sequence and Outputs a Protein

Building a Turing Machine that Computes the Genetic Translation of a DNA Sequence and Outputs a Protein

BS-CS-27-132

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This project implements a Turing machine that computes a biological function: translating a DNA sequence over the alphabet $\{A, T, C, G\}$ into a corresponding protein or phenotypic characteristic. The input is a gene encoded as a finite DNA string, and the output is either a named protein (e.g., insulin, haemoglobin) or a trait determined by that gene. The alphabet $\{A, T, C, G\}$ represents the four nucleotide bases—adenine, thymine, cytosine, and guanine—that form the base pairs of the DNA double helix. The tape of a Turing machine closely resembles a single DNA strand, while a two-tape Turing machine mirrors the structure of the DNA double helix. The machine simulates transcription and translation by mapping codons to amino acids and applying rules of gene expression.

Keywords: Turing machine, DNA, protein synthesis, gene expression, codon mapping, molecular biology, computational biology, transcription, translation, DNA double helix, nucleotide bases, theoretical computer science

4 מחשוב קוונטי

מחשוב קוונטי

BS-CS-27-133

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Keywords:

5 אינפורמציה קוונטית עם תופעת סופר-פוזיציה

אינפורמציה קוונטית עם תופעת סופר פוזיציה

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Keywords:

6 שזירת קוונטית

שזירת קוונטית

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Keywords:

7 אלגוריתם של שור לפענוח של RSA ללא מפתח ע"י פענוח קוונטי

אלגוריתם של שור לפענוח של RSA ללא מפתח ע"י פענוח קוונטי

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Keywords:

8 אסטרטגיות מנצחות בשוק ההון, בלק שולס והיפוכים מקריים בשוק

אסטרטגיות מנצחות בשוק ההון, בלק שולס והיפוכים מקריים בשוק

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Keywords:

9 חישוב קומבינציות של גנים כדי לחשב הסתברות למחלה

חישוב קומבינציות של גנים כדי לחשב הסתברות למחלה

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Keywords:

10 אלגוריתם לפתרון משוואת דיפרנציאלית נומרית

אלגוריתם לפתרון משוואת דיפרנציאלית נומרית

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Keywords:

11 בניית LLM - המודל המתמטי של AI

בניית LLM - המודל המתמטי של AI

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Keywords: